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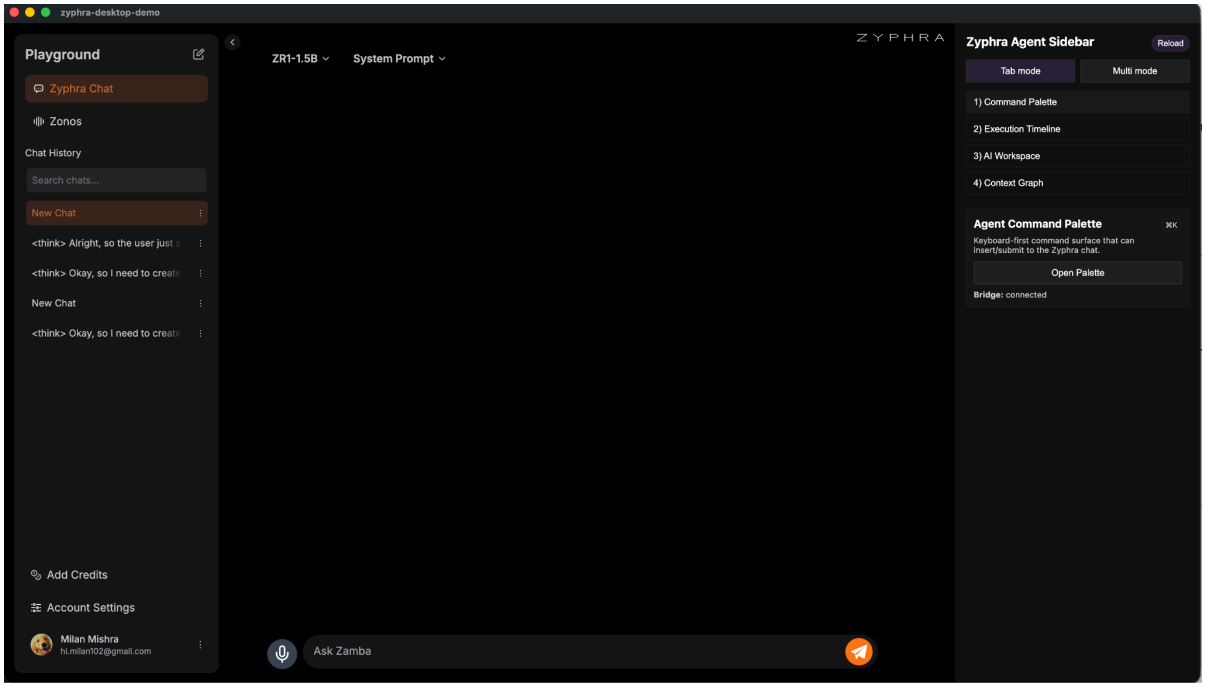
Zyphra Desktop Agent – Demo Overview

This is a React + Electron desktop application prototype showcasing advanced **agent interface thinking**. Built to demonstrate control, context, and explainability in AI-human workflows.

App Layout:

- **Left:** Zyphra chat window
- **Right:** Sidebar with 4 interactive features



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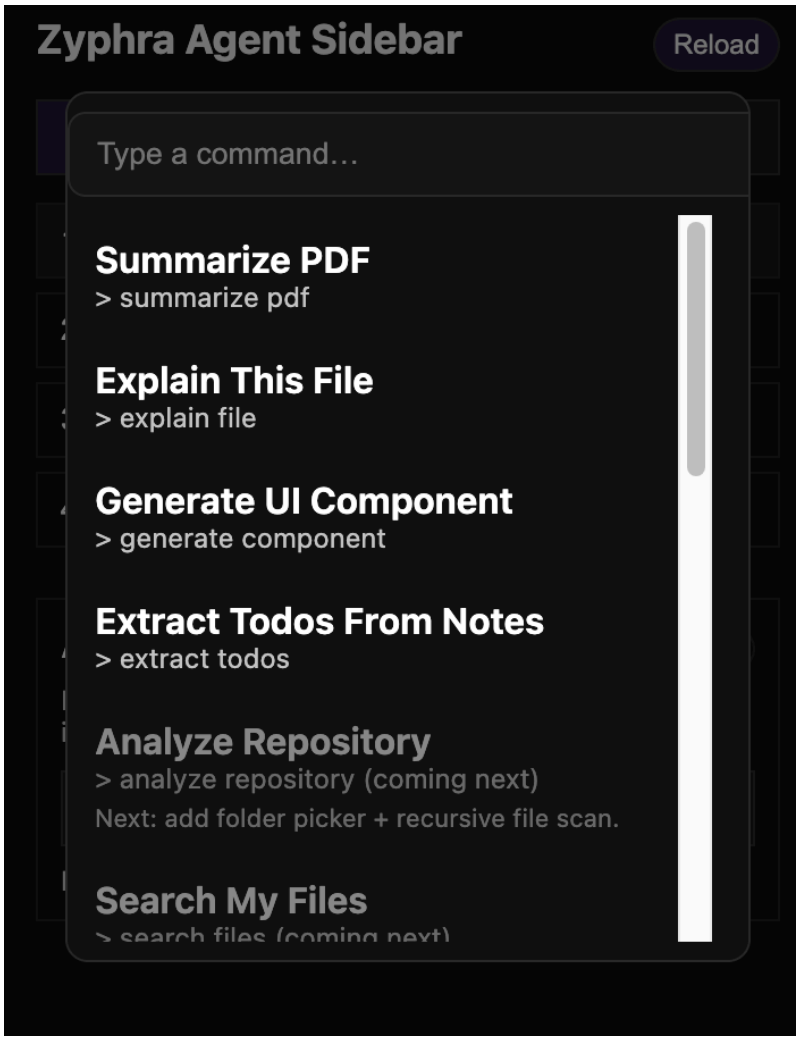
1. Agent Command Palette

Concept: Keyboard-driven command interface (like Raycast/Spotlight) for AI tasks.
Functionality:

- Trigger actions: summarize PDFs, generate UI components, extract todos, analyze repos
- Displays command input, execution status, results, and agent reasoning trace

Why it matters:

- Shows effectiveness of **AI tool orchestration** and **structured prompts**
- Demonstrates **control surfaces** beyond chat



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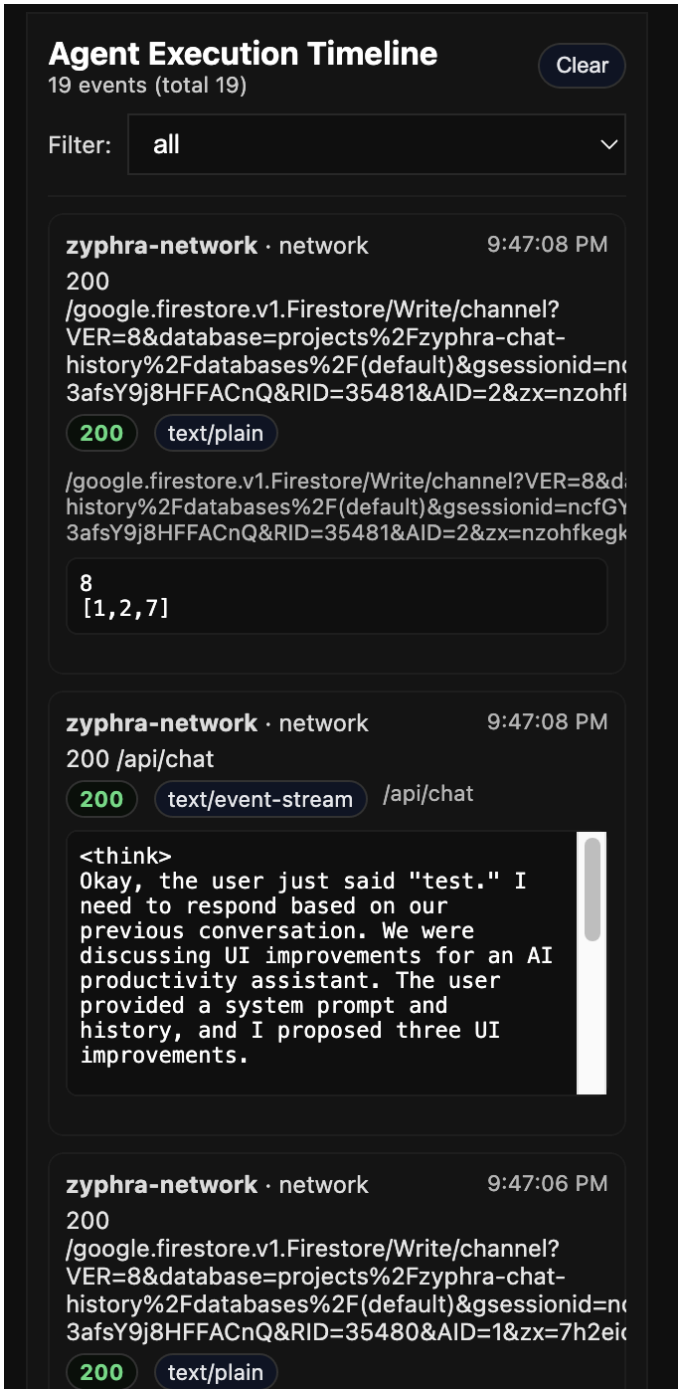
2. Agent Execution Timeline

Concept: Visualizes AI task execution step-by-step
Functionality:

- Timeline of agent actions for any task
- Expandable steps showing input/output at each stage

Why it matters:

- Promotes **trust & transparency** in AI decisions
- Mirrors explainable AI workflows
- Useful for debugging and understanding agent reasoning



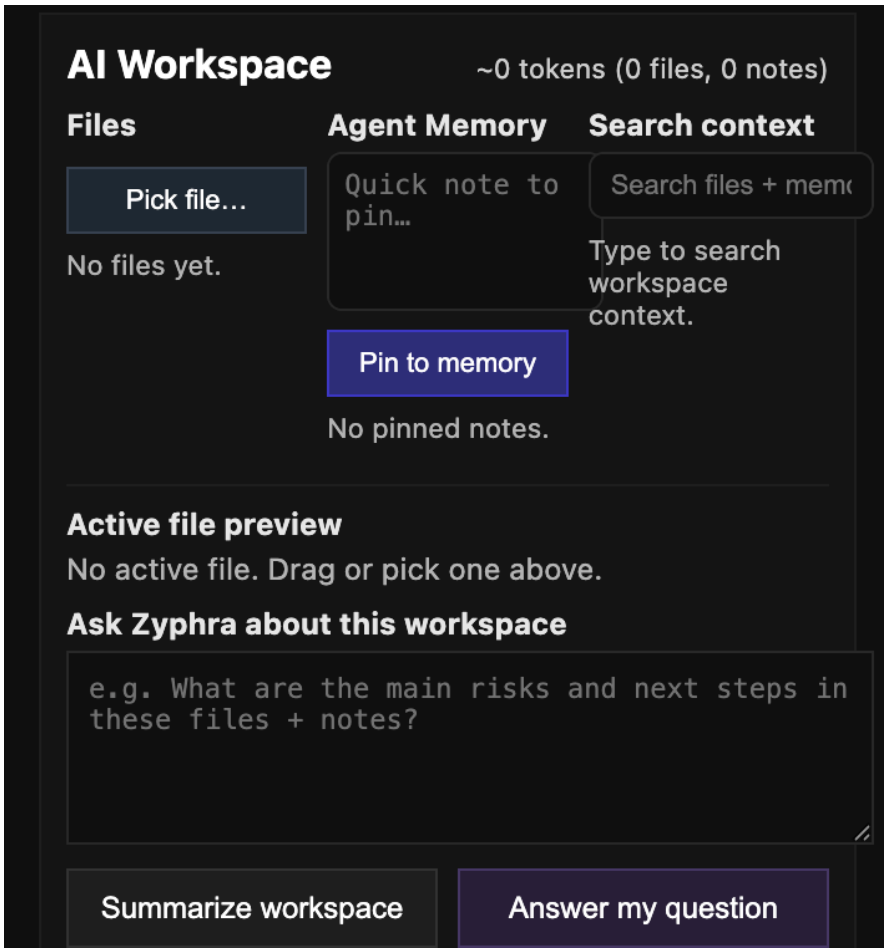
3. AI Workspace

Concept: Multi-panel workspace for AI interactions
Functionality:

- Left: Files
- Middle: Agent Memory
- Right: Search context

Why it matters:

- Moves AI from chat → **productivity workspace**
- Normalizes workflows with **complex AI tools** and **context-driven environments**



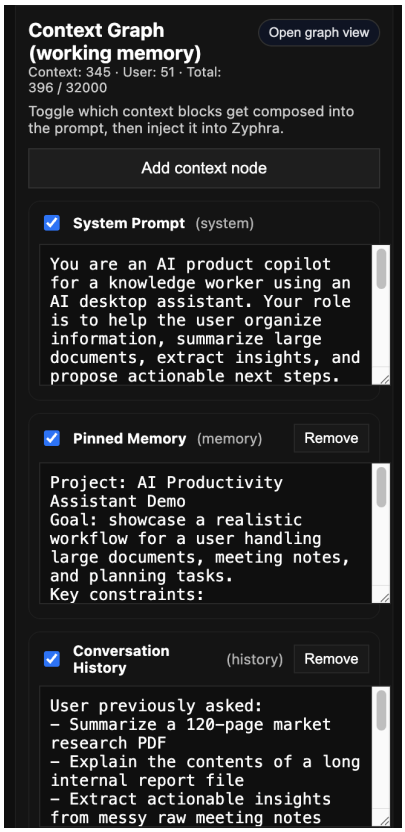
4. Context Graph – AI Working Memory Visualization

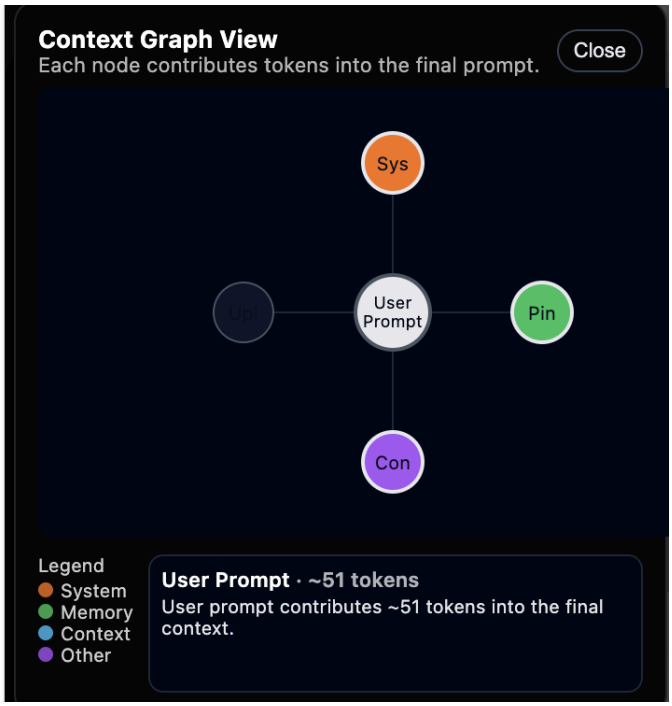
Concept: Visual interface for controlling AI context
Functionality:

- Interactive nodes: conversation, documents, notes, tool outputs
- Toggle include/exclude, pin memory, view token usage
- Prompt preview panel

Why it matters:

- Exposes **hidden context** used by AI
- Encourages mastery of **agent planning, memory, and prompt composition**
- Solves a real-world AI UX problem for model reliability





5. Conclusion & Closing Thoughts

This prototype demonstrates my approach to building **agentic interfaces** that are transparent, interactive, and productivity-focused. It reflects **how users can control AI workflows, understand context, and trust outputs**.

I'm excited to continue exploring these ideas and welcome feedback — this demo is a starting point for **scalable, real-world agent systems**.

There is so much more we can talk about, but I hope this is a good starting point. Zyphra is working on foundational AI technology, and as per my earlier discussion with Jim, has a clear "second-movers advantage". This is quite exciting to me, hence why I did this project.

As an aside, when looking at Zyphra's Zonos API documentation, I was pleasantly surprised to see customizable emotions.This mirrors some experimental work I did a few months earlier on UI/UX focused emergent emotions. The first screenshot below is from Zyphra's API documentation.

Emotion control parameters for the transformer model:

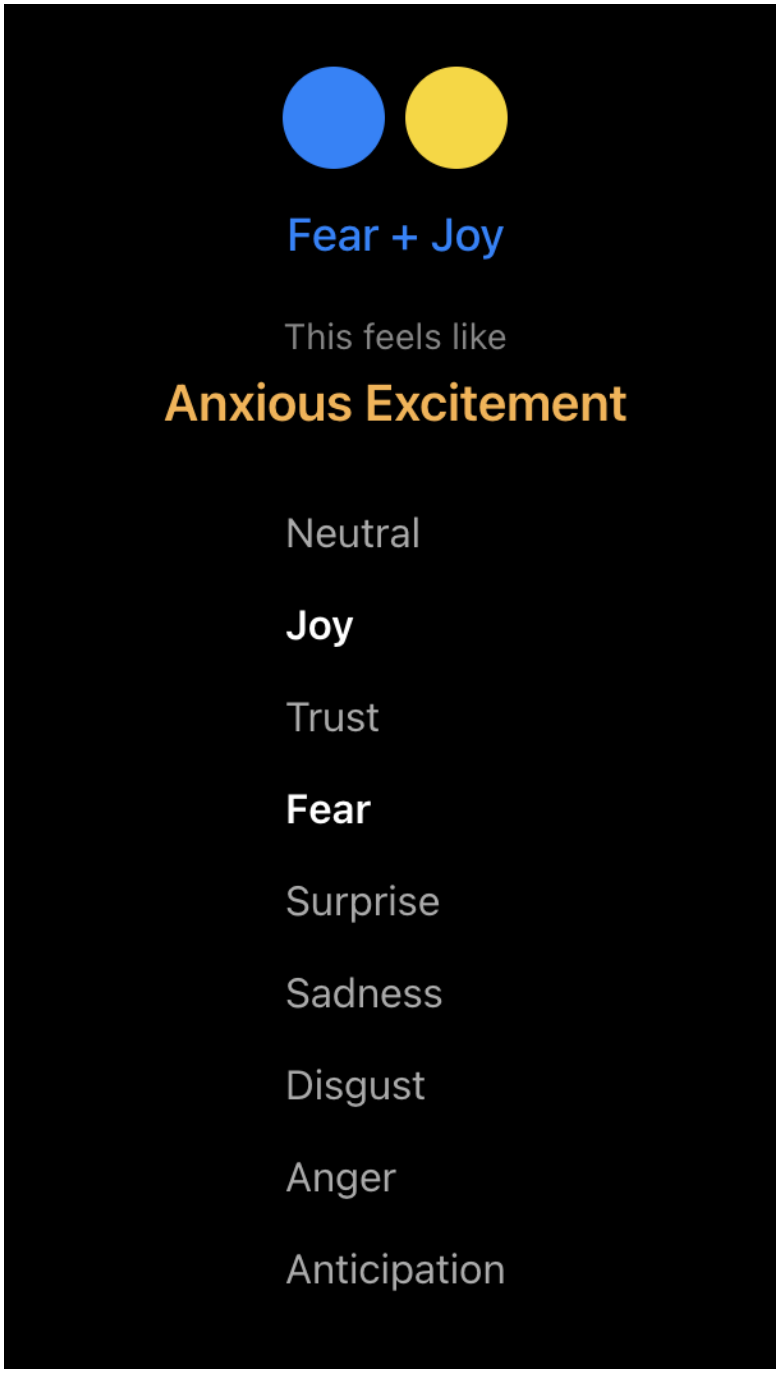
Emotion Parameter	Default Value	Description
`happiness`	0.6	Controls happiness level in voice
`sadness`	0.05	Controls sadness level in voice
`disgust`	0.05	Controls disgust level in voice
`fear`	0.05	Controls fear level in voice
`surprise`	0.05	Controls surprise level in voice
`anger`	0.05	Controls anger level in voice
`other`	0.5	Controls other emotional aspects
`neutral`	0.6	Controls neutrality level in voice

Example with transformer model:

```
# Python example with emotion parameters
audio_data = client.audio.speech.create(
    text="This is a happy message!",
    model="zonos-v0.1-transformer",
    speaking_rate=15,
    emotion={
        "happiness": 0.8,    # Increase happiness
        "neutral": 0.3,      # Decrease neutrality
        "sadness": 0.05,     # Default values
        "disgust": 0.05,
        "fear": 0.05,
        "surprise": 0.05,
        "anger": 0.05,
        "other": 0.5
    },
    pitchStd=50.0           # Slightly higher than default
)
```

Code snippet showing emergent emotions. Again, this is from an experimental web app I was developing a few months ago.

```
function App() {
  // --- Base emotions (UNCHANGED) ---
  const baseEmotions = [
    { name: 'Neutral', color: '#888888' },
    { name: 'Joy', color: '#FFD60A' },
    { name: 'Trust', color: '#30D158' },
    { name: 'Fear', color: '#0A84FF' },
    { name: 'Surprise', color: '#FF9F0A' },
    { name: 'Sadness', color: '#5E5CE6' },
    { name: 'Disgust', color: '#8E8E93' },
    { name: 'Anger', color: '#FF3B30' },
    { name: 'Anticipation', color: '#FFB84C' }
  ]

  // --- Emergent emotion combinations ---
  const emergentByCombination = {
  'Anger+Anticipation': { name: 'Aggressive Ambition', color: '#B22222' },
  'Anger+Disgust': { name: 'Contempt', color: '#8B0000' },
  'Anger+Fear': { name: 'Aggression', color: '#FF0000' },
  'Anger+Joy': { name: 'Frustrated Joy', color: '#FF4500' },
  'Anger+Sadness': { name: 'Bitterness', color: '#4B0082' },
  'Anticipation+Sadness': { name: 'Pessimism', color: '#483D8B' },
  'Anticipation+Disgust': { name: 'Suspicion', color: '#556B2F' },
  'Anticipation+Fear': { name: 'Worry', color: '#1E90FF' },
  'Anticipation+Joy': { name: 'Optimism', color: '#FFD60A' },
  'Anticipation+Surprise': { name: 'Excitement', color: '#FF8C00' },
  'Anticipation+Trust': { name: 'Hope', color: '#FFD700' },
  'Disgust+Fear': { name: 'Aversion', color: '#6B8E23' },
  'Disgust+Joy': { name: 'Wry Amusement', color: '#FFA500' },
  'Disgust+Sadness': { name: 'Remorse', color: '#6A5ACD' },
  'Disgust+Surprise': { name: 'Revulsion', color: '#A52A2A' },
  'Disgust+Trust': { name: 'Moral Outrage', color: '#8B0000' },
  'Fear+Joy': { name: 'Anxious Excitement', color: '#FFB347' },
  'Fear+Sadness': { name: 'Despair', color: '#5E5CE6' },
  'Fear+Surprise': { name: 'Awe', color: '#7D7AFF' },
  'Fear+Trust': { name: 'Submission', color: '#30B0FF' },
  'Joy+Sadness': { name: 'Bittersweet', color: '#FF69B4' },
  'Joy+Surprise': { name: 'Amazement', color: '#FFD700' },
  'Joy+Trust': { name: 'Love', color: '#FF69B4' },
  'Sadness+Disgust': { name: 'Remorse', color: '#6A5ACD' },
  'Sadness+Surprise': { name: 'Shock', color: '#9370DB' },
  'Surprise+Trust': { name: 'Curiosity', color: '#7CFC00' }
}
```

Showing my app running in a web browser. The user selects any combination of 2 emotions, and the emergent emotion appears to them. I believe this could be something to explore in Zyphra's own Zonos model from both a feature standpoint and UI/UX standpoint.